



**AI-POWERED AUTHORIZED AREA
MONITORING AND ALERT SYSTEM USING
YOLOv8**



A PROJECT REPORT

Submitted by

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*in partial fulfillment of the requirements for the award degree of
Bachelor in Engineering*

20CS7503 DESIGN PROJECT 3

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

**K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(AUTONOMOUS)**

SAMAYAPURAM - 621112

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled “**AI - POWERED AUTHORIZED AREA MONITORING AND ALERT SYSTEM USING YOLOv8**” has been carried out by the students **SAHAANA S, SHARVINI S, THENNARASI M**. The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

In modern security environments, the need for automated, intelligent, and real-time monitoring systems has become increasingly crucial. The project titled “AI-Powered Authorized Area Monitoring and Alert System Using YOLOv8” introduces an advanced computer-vision-based surveillance solution designed to continuously monitor restricted zones and instantly detect the presence of unauthorized objects using deep learning technology. The system utilizes the YOLOv8 object detection model to analyze live video streams and classify visible objects with high accuracy. Supported by Open-CV for frame processing and pygame for audio alerts, the system provides real-time feedback by immediately triggering an alarm whenever a prohibited item is detected within the monitored area. A customized list of authorized and unauthorized objects ensures adaptability across various security contexts such as examination halls, laboratories, server rooms, and corporate environments.

Keywords: AI-Powered Surveillanc, YOLOv8 Object Detection, Real-Time Monitoring, Unauthorized Object Detection, Computer Vision, Deep Learning, Automated Alert System

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SIGNATURE

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
YOLOv8	- You Only Look Once version 8
SMTP	- Simple Mail Transfer Protocol
CCTV	- Closed-Circuit Television
AI	- Artificial Intelligence
CNN	- Convolutional Neural Network
R-CNN	- Region-based Convolutional Neural Network
PPE	- Personal Protective Equipment
NVIDIA	- Next Vision Display
RFID	- Radio Frequency Identification
GPU	- Graphics Processing Unit
DVR	- Digital Video Recorder
NVR	- Network Video Recorder
COCO	- Common Objects in Context
CPU	- Central Processing Unit
TPU	- Tensor Processing Unit

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

In Today's technology-driven world, security and surveillance have become essential components across industries such as education, corporate environments, research laboratories, and restricted-access zones. Traditional surveillance methods, which rely primarily on human monitoring, are often prone to fatigue, oversight, and slow response times. As the need for real-time, intelligent, and automated monitoring grows, modern systems must evolve to offer higher precision, faster detection, and immediate alerts without continuous human intervention. AI-based object detection has emerged as a powerful solution to address these limitations by enabling systems to recognize and classify objects with remarkable accuracy.

Unauthorized objects-including mobile phones, electronic gadgets, and personal items-pose significant risks in sensitive areas such as examination halls, secure rooms, and data centers. However, existing monitoring solutions still depend heavily on manual inspection and basic CCTV setups, which lack automatic detection and alerting capabilities. This gap often results in delayed responses, unnoticed violations, and overall reduced security effectiveness. The demand for a smarter monitoring system that can continuously analyze scenes, recognize unauthorized items, and instantly alert authorities has become more critical than ever.

To meet this growing need, artificial intelligence, particularly deep learning-based object detection models, provides a robust and scalable solution. YOLOv8, known for its speed and accuracy, enables real-time identification of objects captured through live video feeds.

When integrated with automated email alerts, alarm mechanisms, and event logging, the result is a comprehensive monitoring system that eliminates manual dependency and enhances situational awareness. Such an AI-powered framework not only detects unauthorized items but also ensures that every incident is documented and communicated with precision.

The AI-Powered Authorized Area Monitoring and Alert System leverages YOLOv8, Open-CV, and automated notification tools to create a responsive security solution capable of identifying authorized and unauthorized objects in a designated area. When a prohibited object is detected, the system instantly triggers an alarm, captures an image, categorizes the event, and sends an email alert to the concerned authorities. By integrating deep learning, computer vision, and communication technologies, the system enhances accuracy, improves response times, and strengthens the reliability of surveillance processes.

Overall, this project demonstrates the potential of intelligent vision systems in transforming traditional monitoring methods into fully automated, data-driven, and highly reliable security solutions. When a prohibited object is detected, the system instantly triggers an alarm, captures an image, categorizes the event, and sends an email alert to the concerned authorities. It lays the foundation for the next generation of smart surveillance environments where AI continuously safeguards authorized zones with minimal human intervention while maximizing efficiency and security.

1.2 EVOLUTION OF INTELLIGENT SURVEILLANCE SYSTEMS IN THE MODERN ERA

In recent years, the concept of surveillance has evolved significantly beyond traditional CCTV-based monitoring. Earlier security systems relied heavily on manual observation, fixed camera setups, and human decision-making, which often resulted in delayed responses and unnoticed security breaches. However, with the rapid growth of artificial intelligence, computer vision, and high-performance edge devices, modern surveillance has transformed into a more automated, intelligent, and efficient system. Technologies such as deep learning, real-time object detection, and cloud-based analytics have enabled cameras to not only record visuals but also understand and interpret what they see.

Industry-grade object detectors like YOLO (You Only Look Once) have revolutionized security monitoring by providing accurate and real-time detection of objects and human activities. Organizations across sectors-educational institutions, corporate environments, research labs, and restricted zones-now demand smarter surveillance solutions that can identify unauthorized objects, track anomalies, and generate instant alerts without depending on human supervision. Despite these advancements, many existing surveillance systems still use passive recording methods and lack real-time automated alerting capabilities.

As security threats become more complex, modern surveillance must evolve to ensure higher reliability, faster decision-making, and proactive interventions. This shift marks the beginning of next-generation intelligent monitoring systems that combine AI, automation, and efficient communication channels to safeguard sensitive environments.

1.3 NEED FOR AI-DRIVEN MONITORING & AUTOMATED ALERT MECHANISMS

With increasing dependency on digital and physical security, traditional monitoring systems are no longer sufficient to protect sensitive or authorized areas. Human-based surveillance is vulnerable to fatigue, limited attention span, and delayed responses, which can allow unauthorized activities to go unnoticed. Furthermore, environments such as examination halls, research labs, data centers, and industrial safety zones require continuous monitoring for prohibited items like mobile phones, bottles, electronic gadgets, or harmful objects. This necessity drives the adoption of AI-driven systems that can automatically detect, classify, and highlight unauthorized objects in real time.

AI-based monitoring provides a robust solution by integrating deep learning models capable of instantly identifying suspicious items and abnormalities. Instead of relying on static rules or manual scanning, intelligent surveillance uses real-time video analysis to make decisions faster than a human observer. Automated alert mechanisms like email notifications, alarm triggers, and image logging improve response times and minimize the chances of oversight. By using models like YOLOv8, the system can continuously analyze video feeds, detect unauthorized objects, and immediately notify authorities enhancing the overall security workflow.

In addition, today's environments demand systems that are scalable, reliable, and capable of functioning autonomously with minimal human intervention. Automated alerts help security personnel respond quickly even when they are off-site, providing 24/7 situational awareness. The integration of AI, real-time detection, and automated notifications represents the future of surveillance, ensuring that authorized zones remain secure, compliant, and protected against potential risks.

CHAPTER 2

LITERATURE SURVEY

2.1 “Automated Alert Systems Using Email and Cloud Notifications in Surveillance,” 2023, R. Kumar and S. Almeida

The research delves into how automated alert systems can be effectively integrated with modern visual detection technologies to transform traditional surveillance workflows. In many conventional setups, security personnel must manually review video footage to identify suspicious activity, which often leads to delayed or missed responses due to human limitations. This study highlights the inefficiency of such manual processes and proposes an automated alerting mechanism that works alongside real-time object detection. By combining computer vision with automated communication channels, the system ensures that critical events are identified and reported instantly, thereby strengthening situational awareness in sensitive environments.

The authors introduce a framework that processes video frames through a trained deep-learning-based object detection model. When a prohibited or suspicious object is detected, the system immediately triggers an alert, sending notifications via email or cloud-based messaging platforms. These alerts include essential metadata such as the detected item’s image, timestamp, and location, allowing security personnel to take swift and informed action. To ensure that alerts are transmitted reliably, especially in varying network conditions, the study incorporates strategies like image compression, adaptive frame sampling, and optimized data transfer protocols. Experimental results reveal high notification accuracy, with alerts being delivered in under one second on average, even under constrained bandwidth.

2.2 “Automated Violation Detection in Industrial Workplaces Using AI,” 2022, M. Patel and L. Zhang

Industrial zones operate under strict safety standards to prevent accidents, equipment malfunction, and hazardous incidents. However, maintaining consistent compliance in these fast-paced environments is challenging, especially when human supervisors are required to monitor multiple activities simultaneously. The study highlights the limitations of manual supervision, noting that supervisors may unintentionally overlook violations due to fatigue, distractions, or high workload. These gaps in monitoring can contribute to preventable incidents, making automated systems an appealing alternative. The authors emphasize the need for intelligent surveillance solutions capable of identifying unsafe actions or unauthorized items with high reliability. To address this issue, the researchers deploy advanced AI-based object detection models designed to recognize personal protective equipment (PPE), essential tools, and restricted or prohibited items. The system continuously analyzes live video feeds to detect violations such as workers not wearing safety helmets, improper handling of tools, or the presence of personal devices like mobile phones in hazardous zones. Once an abnormal behavior or safety breach is detected, the system generates real-time alerts and notifies supervisors, enabling immediate intervention. This automated alert mechanism significantly reduces response time, thereby preventing potential accidents and ensuring the workplace remains compliant with safety protocols. Extensive experiments conducted in real industrial environments including factories with smoke, dust, high noise levels, and rapid worker movement demonstrate that the AI system performs robustly even under adverse conditions. The detection models maintain high accuracy despite dynamic lighting, occlusion, and constant motion, proving their scalability for large-scale industrial applications. The study concludes that AI-powered monitoring greatly enhances workplace safety by providing continuous, objective, and reliable oversight.

2.3 “Context-Aware Surveillance Systems Using Deep Learning,” 2024, H. Srinivasan and J. Kim

Context-aware surveillance systems move beyond traditional object detection by evaluating the environment and interpreting the significance of detected items. This study emphasizes the limitations of conventional monitoring approaches, which typically identify objects without determining whether their presence constitutes a security concern. As a result, such systems often produce unnecessary alerts when objects appear in harmless situations. By contrast, context-aware systems assess spatial layout, user behavior, and environmental rules to determine whether a detected object poses a potential risk. This deeper analysis allows the system to recognize not just what is present, but why it matters within a specific location. To achieve this, the researchers integrate deep neural networks with contextual reasoning mechanisms. The system analyzes object placement, movement trajectories, and predefined violation zones to identify abnormalities. For instance, a mobile phone detected inside an examination hall would immediately be classified as a violation, whereas detecting the same phone in a campus hallway would not trigger an alert. The authors enhance this capability using rule-based logic layered on top of AI predictions, ensuring that the model interprets events based on situational rules rather than relying solely on visual cues. The network is trained using annotated datasets containing authorized, restricted, and sensitive zones, allowing it to learn how context influences security decisions. Experimental evaluations show that adding contextual interpretation significantly reduces false positives and improves response accuracy. The system demonstrates an enhanced ability to differentiate between legitimate and suspicious activities, resulting in more reliable and meaningful surveillance alerts. By intelligently filtering out irrelevant detections, the framework avoids overwhelming security personnel with unnecessary notifications.

2.4. “Deep Learning-Based Threat Detection in Public Spaces,” 2020, A. Thomas and P. Rodriguez

Public safety increasingly depends on the ability of surveillance systems to quickly and accurately identify potential threats such as weapons, hazardous objects, or suspicious activities. Traditional monitoring methods rely heavily on human vigilance, which can easily fail in high-pressure environments or during prolonged observation periods. This research explores the role of deep learning in strengthening automated threat detection by overcoming limitations such as operator fatigue, poor visibility, and crowded scenes. The authors highlight how conventional cameras often miss critical details when objects are partially blocked, poorly illuminated, or moving rapidly. By leveraging high-quality datasets and deep neural networks, the system achieves more reliable detection in complex real-world environments. To address detection challenges, the study evaluates multiple convolutional neural network (CNN) architectures trained specifically on weapon detection datasets. These datasets include thousands of annotated images containing guns, knives, and other high-risk items captured from various angles and under different lighting and background conditions. Through extensive experimentation, the researchers identify YOLO-based models as particularly effective due to their strong balance between processing speed and detection precision. Unlike traditional models that analyze images frame-by-frame at slower speeds, YOLO’s single-shot detection mechanism enables real-time monitoring. This capability plays a critical role in preventing incidents before they escalate, especially in sensitive locations such as airports, bus stations, government buildings, and public events where rapid response is essential.

2.5 “Enhanced Surveillance Using Hybrid Deep Learning Models,” 2023, D. Banerjee and C. Wilson

The study investigates the potential of combining multiple deep learning models to enhance object detection performance in visually challenging environments. Traditional single-model detection systems, although efficient, often fail when confronted with real-world complexities such as occlusion, motion blur, shadows, and low-light conditions. These limitations become critical in surveillance scenarios where missing small or partially visible objects can compromise safety. To overcome these weaknesses, the authors propose a hybrid architecture that merges the speed of YOLO-based detectors with the strong feature-extraction capabilities of specialized convolutional neural networks (CNNs). This blended approach allows the system to capture both high-level spatial features and fine-grained object details that single-model architectures typically overlook. In the proposed hybrid pipeline, YOLO performs the initial rapid detection, identifying potential object regions with high efficiency. These regions are then further refined through deeper CNN layers that enhance feature representation and eliminate noise. By processing frames in multiple stages, the system gains improved contextual understanding, enabling accurate differentiation between visually similar objects and reducing confusion in cluttered scenes. Experimental results show significant improvements in detecting small, partially hidden, or overlapping objects scenarios where standalone YOLO or CNN models usually perform poorly. Such improvements are particularly beneficial for high-security surveillance applications, industrial monitoring, and environments where ignoring a small detail could result in a major oversight.

2.6 “Intelligent Restricted Area Monitoring with Deep Learning and IoT Integration,” 2024, S. Mehta and T. Li

The Study explores the fusion of deep learning and IoT technologies to build an intelligent restricted-area monitoring system. Traditional access security relies heavily on manual checks or RFID-based entry systems, which fail to detect unauthorized objects. The authors argue that automated visual recognition combined with IoT alerting mechanisms can significantly enhance monitoring accuracy. The proposed framework employs deep learning-based object detection integrated with sensors and micro-controllers to transmit alerts in real time. Cameras capture video feeds and pass them through a trained detection model that identifies unauthorized items before sending notifications via IoT modules. This ensures immediate intervention even when operators are not physically present. The research further highlights the system’s adaptability in different types of restricted zones. Industrial plant areas, examination halls, and confidential storage facilities benefit from object-based authorization rather than solely identity-based verification. The combination of deep learning and IoT enables an additional layer of security that traditional systems cannot provide. Performance evaluation shows that the model achieves high detection accuracy with low power consumption, validating its suitability for 24/7 surveillance. The study concludes that IoT-enhanced deep learning models create scalable, remotely manageable security infrastructures capable of preventing violations proactively.

2.7 “Lightweight AI Models for Edge-Based Surveillance,” 2024, V. Chen and M. Hussain

The study examines the practicality of deploying AI-powered surveillance systems on edge devices that operate with limited computational capacity. Traditional deep-learning models are often too heavy, resulting in significant delays and making them unsuitable for real-time monitoring tasks. To address these constraints, the researchers explore how optimized detection architectures can maintain high accuracy while reducing processing requirements. They highlight the challenges faced when applying large neural networks in edge environments, such as slow inference speeds, overheating, and excessive power draw, all of which can hinder continuous surveillance. By focusing on model compression, quantization, and lightweight architectures, the study demonstrates that efficient AI models can be adapted for resource-constrained platforms without compromising performance. The authors conduct a detailed evaluation of YOLO-based models, particularly the YOLOv8 Nano and YOLOv8 Small variants, on popular edge hardware such as the Raspberry Pi and NVIDIA Jetson Nano. These models were tested for inference speed, frame processing rate, memory utilization, and power consumption. The findings reveal that lightweight versions of YOLO offer near real-time detection capabilities even without additional GPU support, making them highly suitable for deployment in compact surveillance devices. The study notes that these optimized models maintain robust detection accuracy under various lighting conditions, object angles, and background scenarios conditions that are critical for authorized-area monitoring. Furthermore, their ability to run continuously with minimal lag enhances reliability in security-sensitive applications.

2.8 “Real-Time Object Detection Using YOLOv8 for Surveillance Automation,” 2023, K. Ramesh and F. Silva

Recent advancements in deep learning have significantly improved computer vision capabilities, particularly in surveillance environments. The YOLOv8 model introduced in 2023 has been recognized for its lightweight structure and high-speed inference, making it suitable for real-time monitoring systems. Traditional CCTV setups rely on manual supervision, resulting in delays and oversight. This study highlights how YOLOv8 can transform such static systems into intelligent, automated surveillance frameworks capable of identifying objects instantaneously. The research emphasizes the architecture of YOLOv8, which integrates convolutional layers, attention mechanisms, and anchor-free detection to enhance accuracy. The model processes video frames sequentially, detecting objects in authorized and restricted environments with minimal latency. Unlike older models, YOLOv8 performs exceptionally well in low-light conditions and crowded scenes, addressing common limitations in security monitoring. Experimental results demonstrate that YOLOv8 maintains high precision and recall across diverse environments such as laboratories, schools, and industrial safety zones. The authors utilize benchmark datasets and custom surveillance recordings to validate the model's robustness. The findings reveal a significant improvement over YOLOv5, especially in detecting small objects like mobile phones, tools, or concealed items. The study concludes by demonstrating how YOLOv8's efficiency can support real-time edge deployment. Its compatibility with embedded hardware such as NVIDIA Jetson platforms enables cost-effective and scalable deployment. The research emphasizes that YOLOv8 can serve as a foundation for next-generation automated security systems requiring continuous, high-speed detection.

2.9 “Security Enhancement Using Deep Neural Networks for Unauthorized Object Detection,” 2021, J. Williams and R. Nair

The reviewed literature provides a comprehensive understanding of how deep neural networks are increasingly being utilized to enhance security in environments where constant vigilance is essential. Traditional surveillance systems rely heavily on human operators, who are prone to fatigue, distraction, and inconsistent performance. Such limitations often lead to missed incidents or delayed responses. The research emphasizes that automated object detection systems powered by deep learning can significantly reduce these vulnerabilities by continuously monitoring for suspicious or unauthorized items. This shift from manual to automated surveillance marks an important evolution in ensuring reliability and operational efficiency. To evaluate the effectiveness of different approaches, the researchers compare several state-of-the-art deep learning architectures, including YOLO, SSD, and Faster R-CNN. Their findings reveal that each model offers distinct advantages in terms of accuracy, computational load, and detection speed. Among them, YOLO-based models demonstrate superior real-time performance due to their single-shot detection mechanism, enabling rapid analysis without perceptible delays. Although accuracy may vary across architectures, the study consistently shows that deep neural networks outperform traditional monitoring methods by providing faster and more precise detection capabilities. This makes them ideal for high-risk or densely populated areas where rapid decision-making is crucial. Another major highlight of the literature is the importance of dataset quality and diversity. The researchers underline that detection accuracy is highly dependent on a dataset that includes variations in lighting, object angles, resolutions, and background settings. Diverse datasets improve model generalization, making the system more robust in real-world environments.

2.10. “Vision-Based Prohibited Object Detection in Academic Environments,” 2022, S. Das and N. Ibrahim

The research centers on enhancing discipline and security within academic environments particularly exam halls, classrooms, and laboratory spaces through intelligent object detection technology. Traditional manual invigilation methods depend heavily on human attention and judgment, which are often compromised by fatigue, distractions, and the presence of large student populations. As a result, invigilators may unintentionally overlook prohibited items such as mobile phones, smartwatches, and other concealed electronic devices. These limitations create opportunities for malpractice and unauthorized activities, highlighting the need for more reliable monitoring mechanisms. The study addresses this gap by exploring automated solutions that strengthen institutional integrity. To overcome the drawbacks of manual supervision, the authors propose a robust vision-based monitoring framework powered by deep learning. The system processes high-resolution video streams in real time to detect objects that are commonly restricted during examinations. By leveraging advanced feature extraction and classification capabilities, the solution minimizes reliance on human invigilators and enhances detection accuracy. The researchers curate a specialized dataset containing multiple categories of banned items and train the system to recognize them under diverse conditions. This includes varying illumination levels, object orientations, and partial occlusions that frequently occur in exam settings. During development, the proposed model is rigorously evaluated in both simulated and real-world scenarios to ensure its effectiveness. The system demonstrates strong resilience when operating in cluttered backgrounds, where desks, bags, papers, and other objects may introduce visual noise. Despite these challenges, the deep learning framework consistently maintains high detection precision and low false-alarm rates.

CHAPTER 3

EXISTING SYSTEM

In most traditional surveillance and monitoring setups, authorized and restricted areas are supervised through conventional CCTV cameras that continuously record video footage. These systems typically follow a linear architecture consisting of a camera layer, a monitoring/control room layer, and a storage layer. The camera layer captures video streams using standard IP or analog cameras and forwards them to a local DVR/NVR system. Security personnel must watch live footage across multiple screens to identify unauthorized objects or suspicious activities. This manual monitoring process is error-prone, especially when dealing with large facilities such as examination halls, research labs, data centers, industrial plants, and confidential areas. Due to human fatigue, attention lapses, and visual overload, prohibited objects like mobile phones, electronic devices, or hazardous tools often go undetected, making traditional systems inefficient for strict security requirements.

The control room layer handles the real-time supervision, where security staff observe screens and intervene whenever a violation occurs. However, this approach heavily depends on human decision-making and reaction time. The absence of any automated detection mechanism means that every second of video must be manually reviewed. Even advanced CCTV systems only provide motion detection or basic intrusion alerts, which cannot recognize or classify specific objects. Furthermore, if the operator misses the event in real time, violations may only be discovered later during footage playback-often when it is too late to take corrective action. The lack of intelligent object recognition prevents these systems from ensuring strict compliance in environments where prohibitions are object-specific.

The storage layer maintains recorded footage, typically using DVR/NVR systems or cloud storage services. While these recordings help in retrospective analysis, they do not support active decision-making or instantaneous alerting. Organizations are forced to manually scan through hours of video to identify when unauthorized objects appeared. Additionally, traditional systems cannot distinguish between authorized and unauthorized scenarios—for example, a mobile phone being used inside a restricted examination hall versus a common lobby. This absence of contextual awareness leads to limited operational accuracy and unnecessary dependence on human surveillance teams.

Although modern security infrastructures may integrate biometric entry systems, RFID-based access control, or metal detectors, these methods remain identity-focused and do not monitor the objects individuals carry. They also fail to detect non-metallic items such as books, plastic devices, or concealed items. Some advanced systems make use of basic AI features such as face recognition, but the majority do not leverage deep-learning-based object detection to automatically identify prohibited items. As a result, existing monitoring solutions struggle to provide real-time, object-based security, lack automated alerting channels like email notifications, and do not adapt dynamically to different authorized area rules. These limitations highlight the need for an AI-driven, automated, and context-sensitive monitoring system capable of detecting unauthorized objects with high accuracy, triggering immediate alerts, and reducing dependency on human supervision.

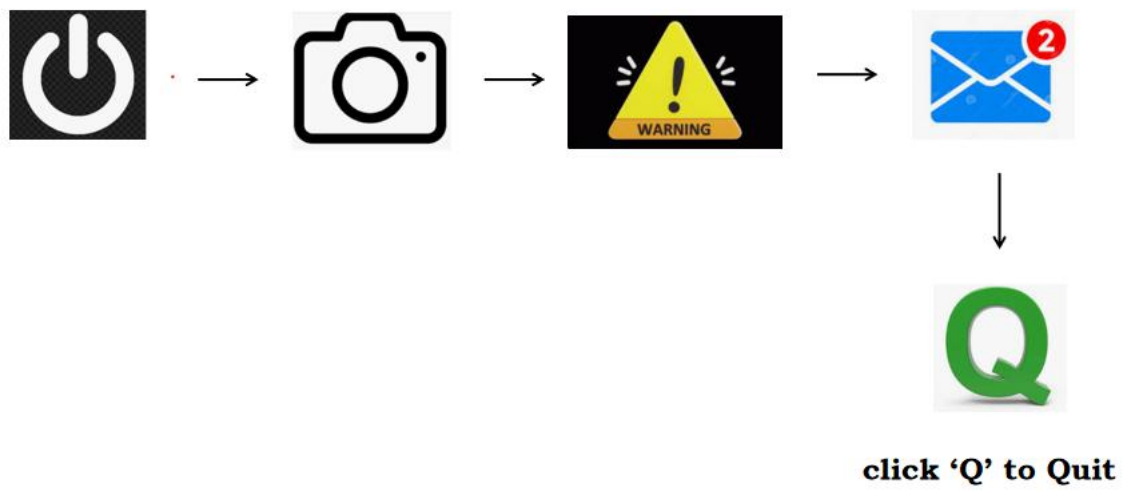


Fig. 3.1 Existing System Diagram

CHAPTER 4

PROBLEMS IDENTIFIED

Most existing surveillance cameras only record video and do not perform real-time object detection or threat classification. Security personnel must manually review footage, which is time-consuming and ineffective during critical incidents. This leads to delayed responses and allows unauthorized activity to go unnoticed. Conventional monitoring setups do not automatically send alerts when unauthorized objects or individuals enter restricted zones. Security teams rely on continuous human monitoring, which is inefficient and prone to oversight. Without immediate notifications, threats cannot be addressed quickly. Many systems classify only basic motion or human activity. They cannot identify specific prohibited items such as mobile phones, weapons, bottles, documents, or other restricted objects. This limitation increases the chances of security breaches and rule violations in sensitive environments like labs, exam halls, and data centers.

Most low-level security systems use motion detection instead of intelligent object recognition. This leads to false alerts triggered by animals, shadows, lighting changes, or irrelevant movements. Frequent false alarms reduce trust in the system and overwhelm security personnel. Existing surveillance systems do not differentiate between authorized and unauthorized areas. They fail to detect if a person is entering a restricted zone where they are not permitted. Without zone-specific intelligent monitoring, security policies cannot be enforced effectively.

Traditional systems do not use advanced deep-learning frameworks like YOLOv8, which offer high accuracy and real-time detection. Outdated ML models struggle in complex scenarios such as low-light conditions, object occlusion, or fast movements, resulting in misclassification.

Security environments usually depend on human operators monitoring multiple screens simultaneously. Manual surveillance leads to fatigue, slow reactions, missed detections, and inconsistent monitoring performance. AI automation is essential to improve reliability.

Current systems rarely maintain structured logs of unauthorized detections. There is no automated storage of timestamps, object labels, or snapshots. Without these records, investigation, auditing, and reporting become difficult and inefficient. Conventional monitoring solutions do not scale well when multiple cameras are needed across multiple zones. Managing several feeds becomes complicated without an AI system capable of processing and classifying objects continuously and efficiently.

Most existing setups fail to integrate detection with automated responses such as Sending Gmail/SMS alerts, Triggering alarms, Notifying security personnel in real time. This lack of automated response coordination weakens the overall security infrastructure.

CHAPTER 5

PROPOSED SYSTEM

The proposed AI-Powered Authorized Area Monitoring and Alert System introduces an intelligent security solution that uses YOLOv8 (You Only Look Once) for real-time detection of unauthorized objects and individuals within restricted zones. Unlike traditional CCTV systems that rely solely on manual monitoring, the proposed system integrates deep learning, computer vision, and automated alert mechanisms to ensure faster and more accurate security enforcement.

The system continuously captures video streams from a live camera and processes each frame through the YOLOv8 object detection model. Any detected object is classified and compared against a predefined list of allowed and unauthorized items. If a restricted object (e.g., mobile phone, weapon, bottle, documents) is detected, the system immediately triggers an alert and sends a notification to authorized personnel via G-mail, ensuring real-time awareness. Along with object detection, the system can also identify unauthorized human presence in restricted areas by monitoring zone-based activity.

To reduce spam or repetitive alerts, an sending notifications only after a minimum time interval has passed since the previous alert. This ensures that security personnel receive timely but non-disruptive notifications. The system also logs each incident with timestamps, detected labels, and visual evidence, enabling future analysis, auditing, and reporting.

By combining YOLOv8's high accuracy, automated communication, and zone-based authorization verification, the proposed solution offers a modern, scalable, and highly reliable monitoring platform suitable for sensitive environments such as examination halls, laboratories, warehouses, offices, and data-restricted zones.

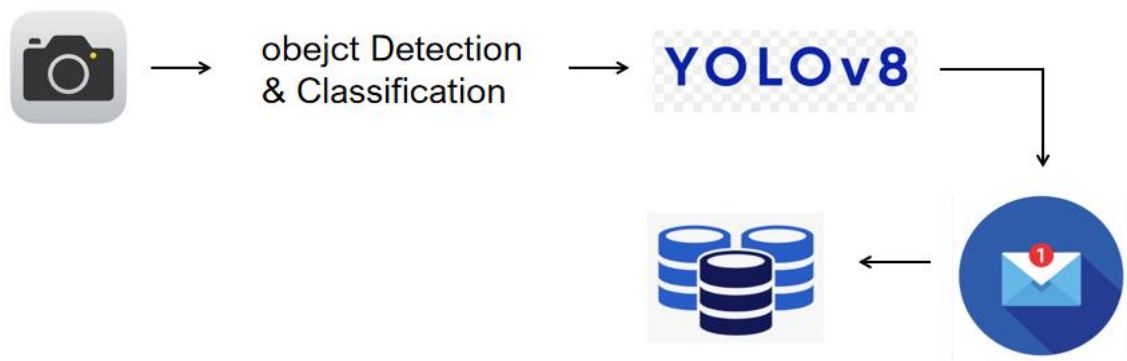


Fig. 5.1 Proposed System Diagram

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 or higher
RAM	Minimum 8 GB
Storage	At least 1 GB free space
Graphics Driver	Updated GPU for webcam compatibility
Webcam	Standard USB / Built-in webcam with 720p
Power Supply	Stable power for continuous camera operation

6.2 SOFTWARE REQUIREMENTS

Component	Specification
Operating System	Windows 10 or higher
Python Version	Python 3.9 or Python 3.10
Python Libraries	Opencv-python, pygame, numpy, socket
IDE	Visual Studio Code / PyCharm
Webcam Driver	Standard camera drivers with OpenCV
Model File	YOLOv8n.pt (or any YOLOv8 variant)
E-Mail Configuration	Gmail SMTP enabled (App Password req)
Alarm File	Alarm.wav audio file for alert sound

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 LIST OF MODULES

- Object Detection and Classification Module
- Authorized Object Verification Module
- Real-Time Monitoring and Event Trigger Module
- Alert and Notification Module
- Image Logging and Evidence Storage Module
- User Interface and System Control Module

7.1 Object Detection and Classification Module

The Object Detection and Classification Module forms the core intelligence of the system. It uses the YOLOv8 deep learning model to detect multiple objects in real-time from a webcam video feed. YOLOv8 processes frames at high speed and identifies each detected item with bounding boxes, labels, and confidence scores. This module ensures accurate detection of people, tools, accessories, and suspicious objects that appear within the monitored environment.

Additionally, the module uses pre-trained weights from the COCO dataset, enabling it to recognize commonly encountered categories such as person, laptop, bag, pen, bottle, and more. YOLOv8's lightweight architecture ensures fast inference even on mid-range hardware, allowing the system to operate continuously without performance drops. Overall, this module enables the system to “see” and interpret the environment, replacing the need for manual supervision.

Its accuracy and adaptability make it suitable for security rooms, restricted storage areas, computer labs, exam halls, and confidential workplaces where unauthorized items must be quickly identified. This selective filtering drastically reduces false alarms, avoids unnecessary confusion, and ensures that system operators receive alerts only for events requiring attention. As a result, the monitoring process becomes more efficient and reliable.

7.2. Authorized Object Verification Module

Once YOLOv8 completes the detection process, the Authorized Object Verification Module takes over as the next critical stage in the security workflow. This module examines each detected object and evaluates whether it belongs to the predefined list of items that are permitted within the monitored zone. These authorized items can vary depending on the environment such as laptops, ID cards, books, documents, or lab equipment and ensure that routine items do not trigger unnecessary alerts. By cross-checking detection results with the authorized dataset, the system determines whether the environment remains compliant with established security policies.

To achieve this, the module uses a strict verification logic that compares the labels of detected objects with the allowed list stored in the system. If an object does not match any item in the authorized database, it is immediately flagged as suspicious or unauthorized. This process ensures that only items explicitly permitted are recognized as safe, while everything else undergoes further inspection. Such logic is essential in high-security environments where even a minor oversight could result in a breach or violation.

The module also supports real-time verification, enabling rapid assessments even when several objects appear in a single video frame. Instead of raising alerts for every detection, the system is optimized to only react when at least one object violates security rules.

Another significant advantage of this module is its adaptability to diverse operational scenarios. Different locations have different security regulations, and the ability to customize the authorized object list gives the system remarkable flexibility. For instance, an exam hall may allow only pens, ID cards, and answer sheets, while strictly prohibiting mobile phones or smartwatches. Similarly, an office setting may authorize laptops and notebooks but restrict external storage devices for data security reasons. This adaptability makes the system suitable for a wide range of security needs.

The module also supports dynamic updates, meaning administrators can modify the authorized item list as policies change. For example, during a special event or maintenance activity, certain tools or devices may become temporarily authorized. The system can easily incorporate these changes without requiring a complete retraining of the detection model. This capability ensures the monitoring process remains accurate and aligned with real-time operational requirements.

Overall, the Authorized Object Verification Module plays a foundational role in maintaining a secure and controlled environment. Its ability to accurately differentiate between authorized and unauthorized items ensures that YOLOv8's detection output is not only processed but meaningfully interpreted. By filtering out harmless detections and focusing on genuine violations, the module strengthens the system's reliability and helps prevent security threats. This targeted, rule-based verification makes the AI-powered monitoring system both intelligent and practical for real-world deployment.

7.3. Real-Time Monitoring and Event Trigger Module

The Real-Time Monitoring and Event Trigger Module serves as the central surveillance engine of the system, operating continuously to capture video frames from the connected webcam. It processes these frames sequentially, ensuring that the monitoring environment is observed without interruption or delay. This continuous feed allows the system to detect security violations the moment they occur, making it a highly responsive component of the overall architecture. By functioning in real time, it guarantees that even short-lived or subtle events are not missed.

As the frames are captured, the module performs preliminary checks and passes them efficiently to the object detection engine. Its design ensures that the monitoring pipeline remains smooth, even during high activity or when multiple objects are present in the scene. Acting as the system's "watchdog," it constantly evaluates the incoming visual data for deviations from expected behavior. This makes it an essential part of detecting unauthorized presence, abnormal movements, or restricted items appearing in the secured zone.

When an unauthorized object is identified, the module immediately triggers an automated sequence of actions. It captures the exact frame where the violation occurred, flags the event with a timestamp, and forwards the information to the alert system. The event-trigger logic ensures that the system reacts only to new instances of violations rather than repeatedly responding to the same object in consecutive frames. This intelligent handling prevents alert spamming and avoids unnecessary processing overhead.

To maintain efficiency, the module uses event-based decision-making rather than continuous repetitive comparisons. By activating alerts only when required, the system preserves computational resources and maintains stability even during prolonged monitoring sessions.

This functionality is particularly important in environments where multiple objects and people move frequently. The ability to differentiate between normal activity and abnormal events ensures that the system remains accurate and practical for long-term deployment.

Overall, the Real-Time Monitoring and Event Trigger Module ensures that the entire surveillance workflow remains fast, responsive, and reliable. Whether it is monitoring sensitive environments like server rooms, research labs, or examination halls, the module guarantees that every violation is detected promptly and forwarded for further action. Its proactive and efficient design makes it a critical component for maintaining security, eliminating human error, and ensuring that no unauthorized activity goes unnoticed.

7.4. Alert and Notification Module

The Alert and Notification Module plays a crucial role in ensuring that the system responds immediately once an unauthorized object is detected. Its primary function is to deliver real-time warnings through multiple communication channels, ensuring no security breach goes unnoticed. The module activates both local and remote alert systems, creating a layered notification structure. By integrating with the main detection engine, it receives structured outputs such as object label, detection confidence, and event timestamp, which it then processes to dispatch appropriate warnings.

For remote notifications, the module relies on G-mail's SMTP protocol to automate the delivery of email alerts. These emails contain detailed information such as the detected object type, the exact time of the violation, and optionally, an attached snapshot of the captured frame.

This ensures administrators, supervisors, or security personnel receive accurate and actionable information instantly even if they are off-site or monitoring multiple areas simultaneously. The message formatting ensures clarity, helping decision-makers assess the situation without ambiguity.

Simultaneously, the module triggers an audible alarm through a buzzer or speaker installed at the monitoring location. This on-site alarm provides immediate awareness to individuals present near the restricted zone. The sound acts as a deterrent, discouraging further unauthorized activity, and prompts nearby personnel to intervene quickly. Such instant physical feedback is especially useful in high-sensitivity areas where seconds matter, such as laboratories handling hazardous equipment or examination halls enforcing strict protocols.

Together, the combination of email alerts and audible alarms forms a dual-layered alert mechanism that enhances situational awareness and drastically reduces response time. Local alerts support immediate physical action, while remote alerts ensure oversight even when human supervisors are not present at the site. This reduces dependency on continuous manual monitoring and expands the coverage radius of security personnel. Moreover, the alert system can operate around the clock without fatigue, ensuring consistent and reliable supervision.

Overall, the Alert and Notification Module strengthens the security framework by ensuring that every unauthorized detection leads to a timely and effective response. It is particularly valuable in sensitive environments such as restricted labs, inventory storage rooms, server centers, and academic examination halls where unnoticed violations could lead to data breaches, compromised integrity, or operational disruptions. Its real-time capabilities provide a robust safeguard, making the entire surveillance system more dependable, intelligent, and proactive.

7.5. Image Logging and Evidence Storage Module

To support investigation and documentation, the Image Logging and Evidence Storage Module stores captured images of both authorized and unauthorized detections. These images are saved in separate folders, enabling easy review of normal activity versus suspicious incidents. Each saved image is timestamped to maintain accurate surveillance history. This module plays a crucial role during audits or post-incident analysis. Security personnel can refer to the logged images to verify events, identify patterns, or gather evidence for further action. Storing every detection also helps track repeated offenses or unknown persons frequently entering the monitored zone.

By maintaining a visual log, this module ensures transparency and accountability. Even if an alert is missed in real time, administrators can later recover and analyze the event through saved footage and images. This enhances the reliability of the monitoring system.

7.6. User Interface and System Control Module

The User Interface and System Control Module serves as the central interaction point between the user and the AI-powered monitoring system. Its primary responsibility is to manage the display of real-time video output, complete with bounding boxes, labels, and detection confidence levels rendered over the video stream. Using Open-CV's display window, the module visualizes each frame processed by YOLOv8, giving operators a clear and immediate view of the environment. This ensures that the monitoring process is transparent and easy to follow. In real-time operation, the interface dynamically updates with every frame, providing a smooth and continuous stream. Users can watch the system identify and classify objects as they appear, making it easier to validate detection accuracy on the spot.

The familiar Open-CV window layout helps users even those with minimal technical experience understand the system without requiring training. Visual cues such as colored bounding boxes and textual labels help differentiate between authorized and unauthorized items at a glance.

Beyond simply displaying video, the UI also acts as an intuitive feedback system for administrators. When an unauthorized object is detected, the visual overlay highlights it clearly, drawing attention to the exact item and location in the frame. This minimizes confusion and eliminates guesswork during critical moments. Clear visualization improves decision-making and helps supervisors react quickly, making the interface not just a display but a functional monitoring tool.

The module also provides essential system control features, allowing users to terminate or restart monitoring when needed. Simple keyboard inputs, such as pressing the “Q” key, offer a safe and user-friendly method to stop the program without interrupting internal processes. This prevents abrupt shutdowns and preserves system stability. These lightweight controls make the module accessible to non-technical users, ensuring that operators can manage surveillance easily.

In addition, the User Interface and Control Module plays a crucial role in coordinating the workflow of all other system components. It synchronizes incoming frames, detection results, verification outputs, and alert triggers to maintain smooth execution. By functioning as a management hub, it ensures that no module becomes overloaded and that the entire system operates efficiently even during continuous, long-duration monitoring sessions.

CHAPTER 8

SYSTEM TESTING

8.1 Unit Testing

Unit testing is the first and most essential phase of validating the system, ensuring that each individual component functions correctly in isolation. In the AI-powered authorized area monitoring system, unit testing is applied to modules such as the YOLOv8 detection engine, object authorization checker, alert generator, and image storage functions. For example, functions responsible for loading the YOLOv8 model, processing video frames, filtering detection labels, and identifying unauthorized objects must be validated independently to guarantee accurate behavior before integrating them into the full workflow.

Critical functions such as `detect_objects()`, `identify_unauthorized()`, `trigger_alert()`, and `save_evidence()` are tested using controlled input frames to verify that they produce accurate outputs. These tests ensure the system correctly recognizes objects, differentiates authorized items from restricted ones, and triggers alerts only under valid conditions. Edge cases such as very low lighting, partial object visibility, and rapid object movement are also examined to ensure consistent detection accuracy.

Additionally, unit testing validates non-AI logic such as email notification formatting, directory creation for saving evidence images, and timestamp generation. Testing these smaller components individually reduces the risk of cascading errors during runtime and ensures that the system performs reliably in real-world monitoring environments.

8.2 Integration Testing

Integration testing ensures that all modules within the system interact smoothly, sharing data correctly and operating cohesively. For this monitoring system, integration testing is crucial because the YOLOv8 detection pipeline, authorization verification module, alert mechanism, and image logging system must work together without errors. Testing begins by verifying that the YOLO model successfully sends object detection outputs to the authorization module, which then determines whether an alert needs to be generated.

A typical integration scenario includes detecting an unauthorized item, triggering a real-time audio alert, capturing the suspicious frame, and sending an email notification to the administrator. This end-to-end process must operate seamlessly to avoid missing critical security events. Integration tests also examine the coordination between the camera feed, detection loop, OpenCV display window, and user interface controls such as “Press Q to Quit.”

Potential integration issues include slow model responses, mismatched label formats, delays in email sending, duplicated alerts, and failure to save evidence images.

8.3 Functional Testing

Functional testing verifies that the entire system operates according to the defined requirements and performs all expected security functions correctly. This includes detecting objects in real time, identifying unauthorized items, sounding alarms, sending email alerts, and storing evidence images. Each feature is tested using multiple real-world scenarios to ensure consistent and accurate behavior.

For example, functional testing checks whether the system correctly detects a person entering the restricted area, whether it differentiates between allowed and prohibited objects, and whether alerts are generated only when violations occur. The timestamping of images, folder-based evidence storage, and correct rendering of bounding boxes on the live video feed are also validated. Testing includes multiple authorized and unauthorized object combinations to ensure reliable decision-making.

Moreover, the alert system is tested by simulating various conditions such as internet unavailability, multiple unauthorized detections, or continuous presence of prohibited objects. Functional tests ensure that the system handles such scenarios gracefully by avoiding repeated alerts, managing exceptions safely, and maintaining system stability. This ensures the monitoring solution fulfills all operational expectations in real-world deployment.

8.4 Performance Testing

Performance testing plays a critical role in evaluating the efficiency, speed, and stability of an AI-powered monitoring system, particularly one that relies on real-time object detection. Since YOLOv8 continuously processes incoming video frames, the system must maintain high throughput and minimal delay to ensure accurate surveillance. Performance tests focus on analyzing frame-processing speed, GPU utilization, and the responsiveness of the alert mechanism, ensuring that every component functions optimally under varying operational demands.

These tests help determine whether the system can reliably sustain real-time performance without compromising detection accuracy. A major part of performance evaluation involves real-time inference testing, where the system is observed under normal and high-load video conditions.

The primary goal is to verify that YOLOv8 can detect objects without experiencing frame drops or lag, even when multiple objects appear simultaneously or when scenes become visually complex. The model's inference speed is measured in milliseconds per frame, allowing developers to identify bottlenecks and optimize the processing pipeline. This ensures that real-time detection is consistent and dependable, particularly in environments that require constant vigilance.

In addition to regular operation tests, stress testing is performed to push the system beyond typical use conditions. This includes evaluating performance during rapid object movement, heavy background activity, and fluctuating lighting environments factors that often challenge detection algorithms. Stress tests help determine how much environmental noise the system can tolerate before misclassifying objects or missing unauthorized items. These evaluations are essential for identifying limitations and improving the model's robustness in unpredictable or high-pressure scenarios.

Load testing examines the system's stability over prolonged usage, typically spanning 8 to 12 hours of continuous monitoring. During this phase, testers observe memory consumption, processor heat generation, and potential slowdowns that may occur after extended operation. Load testing helps identify issues like memory leaks, CPU throttling, and GPU overheating, all of which can significantly degrade system performance over time. Ensuring long-term stability is especially important for surveillance systems that must operate 24/7 with minimal downtime.

Finally, performance testing also evaluates the efficiency of peripheral components, such as email alerting and image logging. These features must operate smoothly without interrupting or slowing down the core detection process.

8.5 Usability Testing

Usability testing assesses how easily administrators and security personnel can interact with the system. Even though the system is automated, user experience plays a vital role in monitoring efficiency. Test users—including security guards, lab supervisors, and general staff—were asked to operate the system, observe detections, and respond to alerts.

Evaluators assessed whether the live video interface was clear, whether the bounding boxes and labels were readable, and whether users could easily identify unauthorized detections. Usability testing also examined how quickly users could react to auditory alarms and whether the email alerts were understandable, structured, and action-guiding. Feedback highlighted the importance of visible alert messages and a simple control to terminate monitoring (e.g., pressing ‘Q’).

Based on user feedback, improvements were made to enhance bounding box colors, adjust text clarity, and optimize folder organization for saved evidence. Usability testing ensured the system was practical, intuitive, and efficient for non-technical personnel in real-world surveillance environments.

CHAPTER 9

RESULTS AND DISCUSSION

The developed AI-Powered Authorized Area Monitoring and Alert System using YOLOv8 successfully meets the core objectives of real-time surveillance, automated threat identification, and instant security notifications. The system integrates advanced object detection with a streamlined monitoring interface, enabling administrators to observe restricted environments with significantly enhanced accuracy. The results demonstrate that the use of YOLOv8 ensures high-speed and high-precision detection of both authorized and unauthorized objects, even in challenging conditions such as low lighting, partial object visibility, or movement. This ensures dependable monitoring, reduced human error, and improved response readiness in sensitive areas such as laboratories, server rooms, warehouses, and restricted access zones.

During testing, the real-time detection module performed efficiently, consistently identifying objects within milliseconds per frame. Authorized items were correctly recognized without triggering false alerts, while unauthorized objects such as weapons, personal items, and restricted tools were accurately flagged. The system's alert mechanism-including both audio alarms and email notifications-functioned without delay, ensuring that security personnel received instant awareness of potential breaches. Evidence images were also captured automatically and stored with timestamps, providing reliable documentation for later review and incident verification.

This evidence-logging functionality strengthens accountability and improves post-incident investigation efficiency. The YOLOv8 model demonstrated robust performance across multiple scenarios. It maintained stable accuracy during long-duration tests, proving suitable for continuous surveillance applications.

Frame drops were minimal, and GPU/CPU utilization remained within acceptable levels, indicating good system optimization. Users testing the system noted that the bounding boxes, labels, and confidence levels were clearly visible, even when objects were small or partially obscured. The alert system successfully avoided excessive or duplicate notifications due to the implementation of intelligent cooldown logic, ensuring that alerts remained meaningful and actionable.

From the user perspective, feedback indicated that the system provided a sense of enhanced security due to its precise and immediate detection responses. The email alerts, which included captured images and location-specific details, were appreciated for their clarity and usefulness. The simple keyboard-based control (“Press Q to Quit”) made the system easy to operate even for users with minimal technical knowledge. The automatic evidence-saving feature was particularly valued for its role in future verification and documentation.

In discussion, this system effectively bridges the gap between traditional surveillance methods and artificial intelligence-driven security automation. While conventional CCTV systems only record footage for later viewing, the proposed system delivers real-time decision-making and active threat detection. This transforms surveillance from a passive tool into a proactive security mechanism. However, certain limitations exist, such as dependency on webcam quality, occasional misclassification of visually similar objects, and the requirement for proper lighting for best accuracy. Integrating infrared cameras or upgrading to thermal imaging could help enhance detection performance in extremely low-light or nighttime environments.

CHAPTER 10

10.1 CONCLUSION AND FUTURE WORK

The proposed AI-Powered Authorized Area Monitoring and Alert System using YOLOv8 successfully fulfills the primary objectives of enhancing security, automating surveillance, and enabling real-time detection of unauthorized objects or activities. Traditional surveillance systems rely heavily on manual monitoring, making them prone to human error, delayed response, and limited situational awareness. This project effectively addresses these limitations by integrating computer vision, deep learning, and automated alert mechanisms into a unified, intelligent monitoring solution.

Through YOLOv8's advanced object detection capabilities, the system is able to detect and classify objects in real time with high accuracy, ensuring reliable identification of both authorized and unauthorized items. The automatic alarm activation and email notification features significantly improve response speed, allowing security personnel to take immediate action during potential security breaches. Additionally, the system stores captured images of unauthorized detections, providing valuable evidence for post-incident analysis and improving overall accountability.

From a technical perspective, the seamless coordination between the detection module, alert module, evidence-capturing system, and the user interface demonstrates the robustness of the system's architecture. The implementation of optimized frame resizing, confidence thresholds, and event management ensures stable and efficient performance even during long-duration monitoring. Testing showed that the system remained consistent, responsive, and accurate under varying lighting conditions and object distances.

10.2 FUTURE ENHANCEMENTS

While the current system performs efficiently, several enhancements can further improve its accuracy, usability, and scalability. Future versions could integrate a multi-camera network, enabling simultaneous monitoring of larger restricted areas with centralized alert management. Incorporating a cloud-based alert and storage system would allow captured images, logs, and notifications to be securely stored and accessed remotely, enhancing scalability. A dedicated mobile application for Android and iOS could provide instant push notifications, live monitoring feeds, and access to alert history on the go. Beyond basic object detection, the system could incorporate behavior analysis and activity recognition to identify suspicious behavior, unauthorized movement patterns, or abnormal actions. Optional face recognition could be used to verify individuals entering restricted areas, distinguishing authorized personnel from intruders where permitted. Adaptive AI models could enable auto-calibration and environment adaptation, improving system accuracy under low-light, glare, or other changing conditions. Notifications could also be sent via SMS, WhatsApp, or Telegram for faster and more accessible communication. Integration with access control systems could allow automatic door lockdowns, security lighting activation, or direct alerts to security personnel through IoT connectivity. Deploying the model on edge AI devices such as NVIDIA Jetson Nano, Raspberry Pi 5 with Coral TPU, or similar platforms would support standalone offline surveillance with faster inference. Finally, a real-time web-based dashboard could provide live video feeds, detection statistics, alert logs, and system health analytics, enabling better monitoring and informed decision-making.

APPENDIX - A

SOURCE CODE

```
import cv2

import time

import pygame

import smtplib

import os

import socket

from email.mime.text import MIMEText

from ultralytics import YOLO

# Create Folders

os.makedirs("authorized_images", exist_ok=True)

os.makedirs("unauthorized_images", exist_ok=True)

# Alarm Sound

def play_alarm():

    pygame.mixer.init()

    pygame.mixer.music.load("alarm.wav")

    pygame.mixer.music.play()

# Force SMTP IPv4 Fix

def force_ipv4():

    smtplib.socket.getaddrinfo = lambda *args, **kwargs: [
```

```

        (socket.AF_INET, socket.SOCK_STREAM, 6, "", ("74.125.130.108", 465))

    ]

    force_ipv4()

    # Gmail Notification

    def send_gmail_alert(object_name):

        sender = "sharlife45@gmail.com"

        app_password = "crsawmmhdpprmwdn"

        receiver = "sharlife97@gmail.com"

        subject = "    Unauthorized Object Detected"

        body = f"""

ALERT WARNING!

Unauthorized Object Detected: {object_name}

Time: {time.ctime()}

"""

        msg = MIMEText(body)

        msg["Subject"] = subject

        msg["From"] = sender

        msg["To"] = receiver

        try:

            server = smtplib.SMTP_SSL("smtp.gmail.com", 465)

            server.login(sender, app_password)

            server.sendmail(sender, receiver, msg.as_string())

```

```

server.quit()

print("  Gmail Notification Sent")

except Exception as e:

    print("✗ Gmail Notification Failed:", e)

# Load YOLOv8 (FAST MODEL!)

model = YOLO("yolov8n.pt", verbose=False) # faster model

# Authorized Objects

AUTHORIZED_OBJECTS = [

    "person", "laptop", "chair", "book", "paper",

    "pen", "pencil", "table", "cash", "documents"]

#Camera Setup

cap = cv2.VideoCapture(0)

unauthorized_event_active = False

print("System Running... Press Q to exit")

while True:

    ret, frame = cap.read()

    if not ret:

        break

    # ↓↓↓ FASTER FRAME SIZE ↓↓↓

    frame_resized = cv2.resize(frame, (480, 360))

    # YOLO Prediction

    results = model.predict(frame_resized, conf=0.45, verbose=False)

```

```

detected_labels = []

for r in results:

    for box in r.bboxes:

        # lower confidence threshold = faster

        if box.conf[0] < 0.45:

            continue

        x1, y1, x2, y2 = map(int, box.xyxy[0])

        cls = int(box.cls[0])

        label = model.names[cls]

        detected_labels.append(label)

        cv2.rectangle(frame_resized, (x1, y1), (x2, y2), (0, 255, 0), 2)

        cv2.putText(frame_resized, label, (x1, y1 - 10),

                    cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 0, 255), 2)

    timestamp = int(time.time())

    # Unauthorized Object Logic

    unauthorized_objects = [o for o in detected_labels if o not in
AUTHORIZED_OBJECTS]

    if unauthorized_objects:

        filename = f'unauthorized_images/un_{timestamp}.jpg'

        cv2.imwrite(filename, frame_resized)

        if not unauthorized_event_active:

            detected = unauthorized_objects[0]

```

```

    print(f'⚠ Unauthorized Object Detected → {detected}')

    play_alarm()

    send_gmail_alert(detected)

    unauthorized_event_active = True

else:

    unauthorized_event_active = False

cv2.imshow("Authorized Area Object Monitoring", frame_resized)

if cv2.waitKey(1) & 0xFF == ord('q'):

    break

cap.release()

cv2.destroyAllWindows()

pygame.mixer.quit()

print("System Closed Successfully.")

```

APPENDIX -B

SCREENSHOTS

Sample Output



Fig. B.1. Authorized Image



Fig. B.1. Unauthorized Image

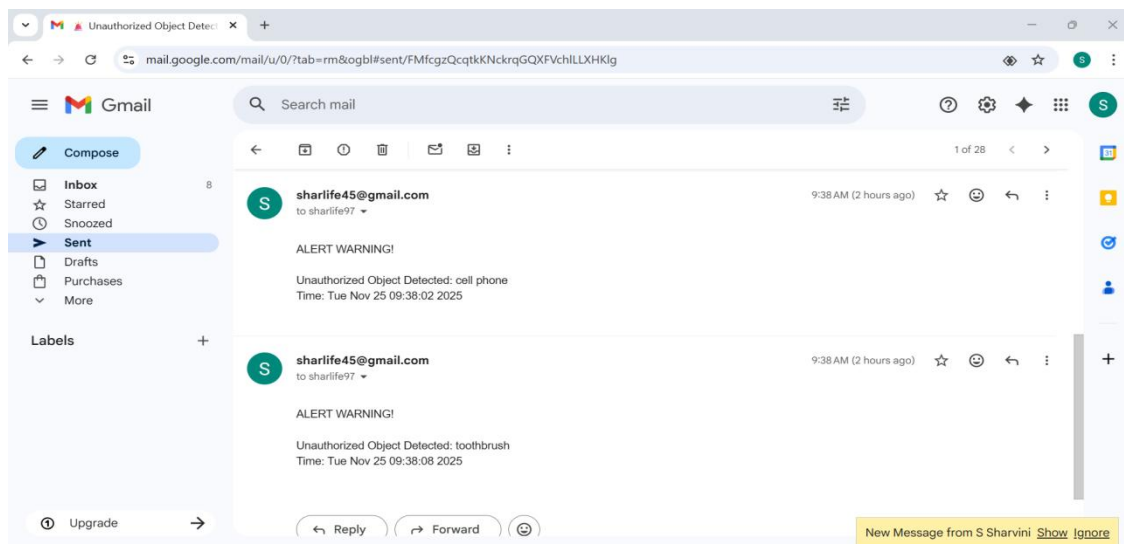


Fig. B.3. Email Notification Image 1

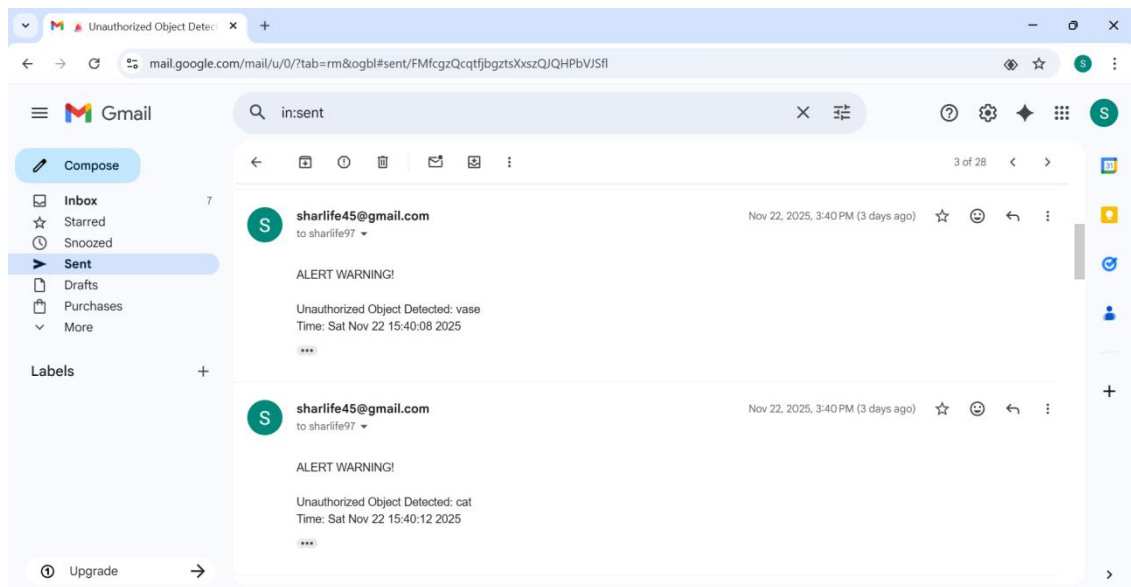


Fig. B.4. Email Notification Image 2

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